
Dynamic First-Crossing Behavior in Kalshi Sports Markets: Evidence of Longshot Bias and Calibration Dynamics

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Project Title: Dynamic First-Crossing Behavior in Kalshi Sports Markets

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1 Problem Statement

Kalshi, the most popular prediction market in the United States, is regulated as a Designated Contract Market allowing it to function in areas where traditional gambling or sports betting may be more heavily regulated. Each contract trades between \$0-\$1 and pays \$1 if YES, \$0 if NO. Despite this legal distinction, the highest volumes of trades occur in the sports category with Kalshi facilitating more than 1.5 billion dollars of sports outcome trading in 2025. This makes sports one of the most interesting domains for evaluating how well prediction markets aggregate information.

Prediction markets are often used as real-time indicators of how people collectively assess the likelihood of future events. These markets should accurately reflect all available information efficiently, making them useful for both forecasting event probabilities and studying belief formation. One aspect of this framework that is central to how Kalshi prediction markets aggregate information is that markets on Kalshi are symmetrical. This symmetrical market setup means that anytime a trader is making a trade at a given price, another individual trader is taking the other side of that trade making trading price a consensus between multiple market participants on the probability of the event outcome. These agreed upon probabilities at the scale of an entire market should mean that prices reflect the different information and beliefs that all market participants have, weighted by the conviction demonstrated in how many contracts they are willing to buy.

Based on the symmetrical consensus Kalshi probabilities should be well calibrated: Events trading at 20 cents should occur 20% of the time, and events trading at 80 cents should occur 80% of the time. However, calibration in probability tails is challenging due to behavioral biases, liquidity constraints and thin participation. Our project aims to interrogate whether those probabilities are well calibrated at the extremes. Evaluating prediction markets only by their final prices can miss important dynamics that occur during trading. A market may temporarily become extremely polar and later recover, or it may remain low until expiration, and those different paths may contain meaningful information about market behavior.

This project studies Kalshi sports markets using a first-crossing framework. Instead of focusing only on terminal prices or final outcomes, the analysis examines whether and when each market's price falls below a low threshold of 5 cents. This approach allows us to investigate whether market odds touching an extremely low probability during the course of a market being traded has predictive power on the final market outcome. The goal is to understand whether low-price crossing is associated with eventual resolution outcome and whether this behavior differs across contracts that resolve YES versus NO. In doing so, the project evaluates longshot behavior and calibration dynamics at the level of the full price path.

2 Methodology

The main evaluation strategy is a first-crossing threshold analysis. For each market, the price path was examined to determine whether the bid price ever fell below 5 cents before the market resolved. If that 5 cent threshold was crossed, the market was marked as a positive crossing case. These positive markets were the main subset of all markets this analysis focused on. The analysis also recorded the minimum observed bid price and the relative position of the crossing within the price path for later calibration bin analysis.

This approach treats threshold crossing as a dynamic measure of market pessimism. Rather than asking only whether a contract ended up accurately priced at expiration, the method asks whether the market temporarily became extremely cheap during trading. The resulting dataset was then compared across final outcomes to determine whether markets that resolved NO crossed below the threshold more often than markets that resolved YES.

The analysis also includes calibration-style evaluation. Markets were grouped into bins based on the first observed price below 5 cents, and within each bin the average predicted probability was compared with the actual YES rate. Bootstrap confidence intervals were computed to assess the uncertainty in each bin,

especially in the sparsest longshot ranges. This provides a more standard forecast-evaluation perspective in addition to the path-based first-crossing analysis.

2.1 First-Crossing Definition

A market was considered to have crossed below 5 cents if its bid price ever reached a value less than or equal to 0.05 before the market resolved. The minimum bid across the observed price path was used as the main price summary. This allowed the analysis to capture both a binary threshold-crossing event and a continuous low-price measure.

2.2 Calibration Bins

To examine probability accuracy, the low-price region was divided into bins: 0–1%, 1–2%, 2–3%, 3–4%, and 4–5%. For each bin, we computed the average predicted probability and the observed YES rate. We then used bootstrap resampling with replacement to construct 95% confidence intervals for the observed YES rate. This approach was chosen to better account for the non-normal distribution of outcomes in these extreme low-probability ranges.

3 Data Collection and Preprocessing

The data were collected from the Kalshi API using a custom Python scraper interrogating all sports markets traded during 2025. Series on Kalshi are a collection of markets which represent different categories of events markets eg. NBA or NFL markets. The scraper first gathered information on all sports series during our year of interest, then pulled market-level metadata. The raw market metadata were saved for each market, and filtered versions were created for markets that were resolved during our year of interest and suitable for analysis. For each of those suitable markets, market trading histories were then collected in the Kalshi candlestick format. Candlesticks are a way of representing trading data for a period of time and, relevant to this analysis, Kalshi candlesticks consist of a high and low bid, ask, and price for that time period as well as the volume of transactions during that candlestick. Our analysis focused on 1 hour long candlesticks because that time resolution balanced information granularity against the number of API requests needed to collect the information. Individual candlestick files were stored separately by market ticker in one directory.

The full raw market dataset contains 304,252 markets, of which 289,845 were finalized and used for further analysis. After applying a high-volume filter, the analysis focused on 79,345 markets. The final first-crossing dataset contains 1,442 candle files with usable price paths which is more than enough data to conduct our testing. Of these final markets, 1,242 markets crossed below 5%, which formed the main sample for the first-crossing analysis.

The outcome breakdown shows that 355 of the markets crossing the 5% odds threshold resolved NO and 263 of those markets resolved YES. These counts show that threshold crossing is common in both outcome groups, but more frequent among markets that resolve NO. The calibration bin analysis used the same candle-derived price summaries and outcome labels to evaluate whether the implied probabilities were aligned with realized outcomes.

Table 1: Summary of project dataset sizes.

Quantity	Value
Raw markets scraped	304,252
Finalized markets used	289,845
High-volume markets used	79,345
Candle files analyzed	1,442
Markets crossed below 5%	1,242
Crossing markets resolving NO	355
Crossing markets resolving YES	263

4 Results and Evaluation

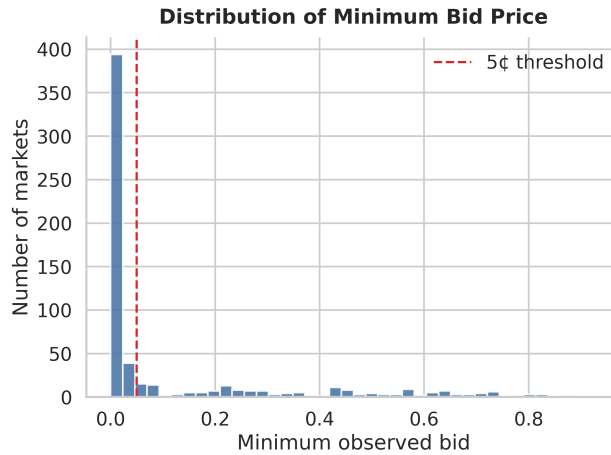


Figure 1: The minimum bid distribution is concentrated near zero, indicating strong longshot behavior.

This analysis found that the distribution of minimum bid prices is heavily concentrated near zero. This means that many markets spent at least part of their trading life in an extreme longshot range. That pattern is consistent with the idea that a substantial fraction of contracts become very cheap before resolution, even if they later recover or do not ultimately resolve in the expected direction. This concentration is not necessarily unexpected because as the event resolution draws closer it is reasonable for market certainty to go up leading to the minimum trading price to trade at low prices towards the end of their life. It is interesting though that traders are still willing to trade on events where markets price the outcomes as so certain. This could be indicative of a high market appetite for longshot odds.

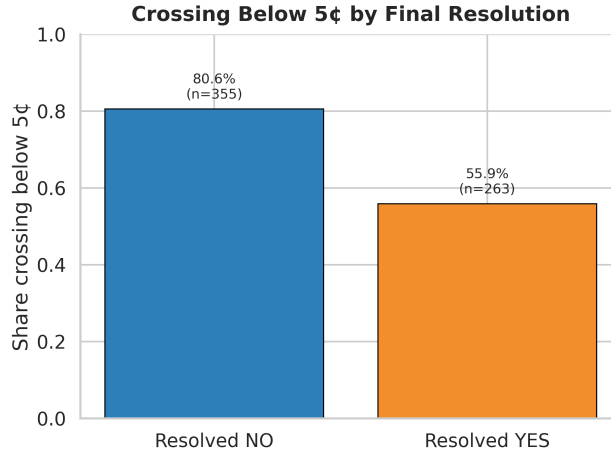


Figure 2: Markets resolving NO crossed below 5 cents more frequently than markets resolving YES.

The distribution of outcomes in the set of markets which crossed the 5 cent threshold show a clear asymmetry in threshold-crossing behavior. Markets that ultimately resolved NO crossed below 5% more often than markets that resolved YES. In the current sample, 355 crossing markets resolved NO and 263 crossing markets resolved YES. This difference suggests that low prices are not evenly distributed across outcomes and that the threshold-crossing event carries information about final resolution.

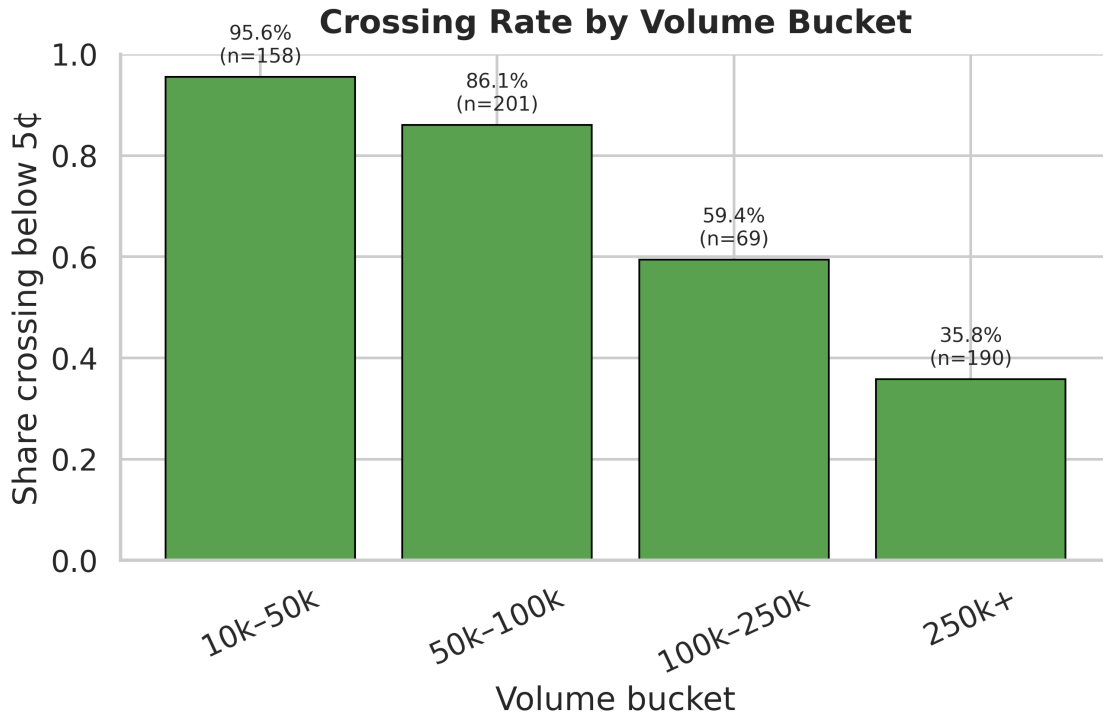


Figure 3: Crossing behavior varies across volume buckets, suggesting that liquidity matters.

We chose to bin markets by volume to investigate if there is a correlation between trading volume and market polarity. Higher-volume markets tend to provide cleaner price paths and may be more informative than thinly traded contracts. In addition, lower trading volume in markets makes them more susceptible

to volatility from relatively large trades being placed. More thinly traded markets seem to cross the 5 cent threshold more often than higher volume markets, indicating that The crossing rates across volume buckets are summarized in the figure and appendix tables, and they show that liquidity should be considered when interpreting threshold-crossing behavior.

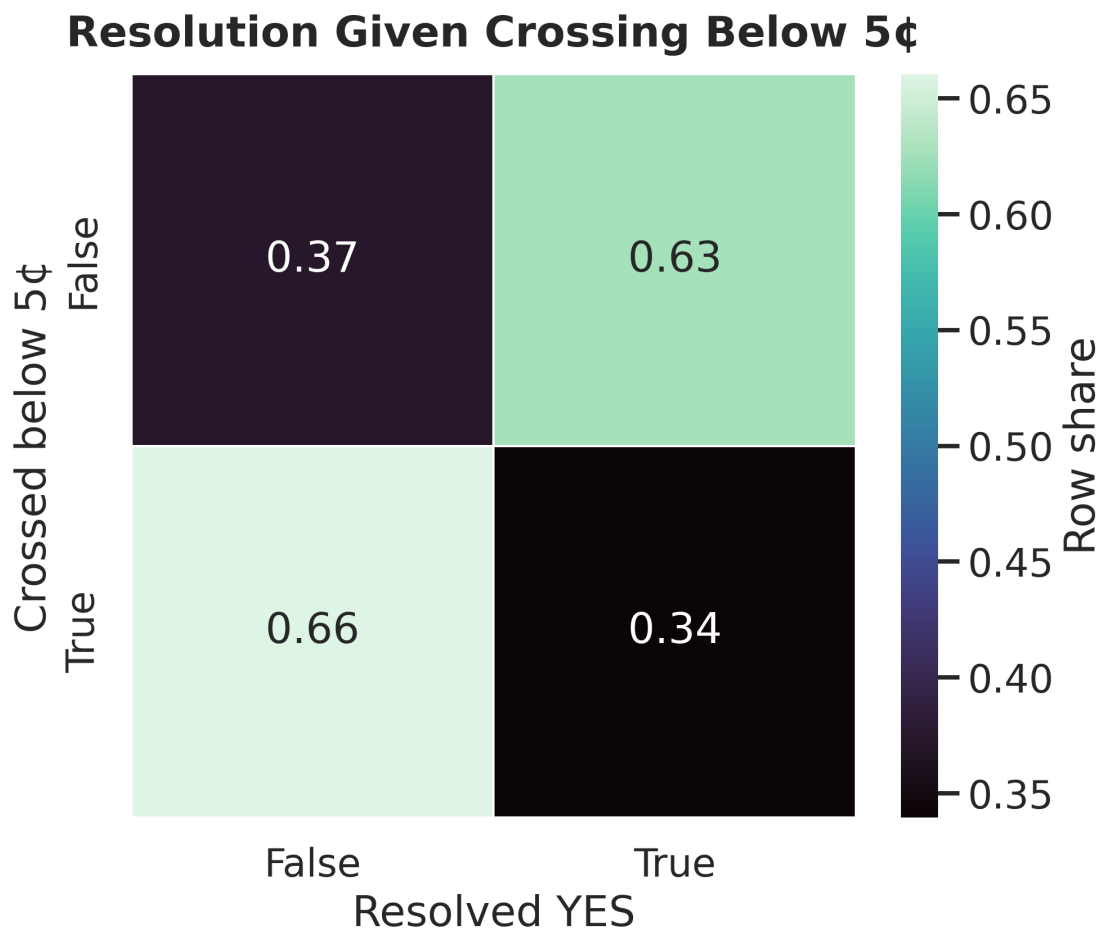


Figure 4: Markets that crossed below 5 cents still resolved both YES and NO, showing that threshold crossing is informative but not deterministic.

The conditional resolution analysis further shows that crossing below 5% does not guarantee a NO final outcome. Even among markets that crossed the low threshold, many still resolved YES. This means the threshold crossing is informative but not deterministic. It provides a signal of market dynamics, not a perfect prediction rule.

4.1 Calibration Analysis

To better evaluate probability accuracy, we grouped markets into bins based on the first observed price below 5% and compared the average predicted probability to the actual YES rate in each bin. We then bootstrapped each bin 1,000 times to produce 95% confidence intervals. This analysis shows that low-probability bins are not perfectly calibrated and that uncertainty becomes larger in the sparsest bins.

The calibration plot compares the average predicted probability on the x-axis with the observed YES rate on the y-axis. A diagonal line indicates perfect calibration. Deviations from the diagonal show

where the market appears to overstate or understate the probability of the event. The bin table and bootstrap intervals make it clear that longshot bins are noisy, but the overall pattern is still informative.

5 Evaluation Strategy

The primary evaluation strategy is first-crossing analysis at a 5-cent threshold. This strategy evaluates the market based on whether the bid price ever falls below the threshold before resolution and compares that behavior across final outcomes. It is a path-based evaluation method, which means it considers the entire price trajectory rather than only the final observed price.

To strengthen the evaluation, the project also includes bin-based calibration analysis and bootstrap confidence intervals. A separate calibration plot compares average predicted probability to observed YES rates, while a diagonal $y = x$ line indicates perfect calibration. These strategies assess whether the market's implied probabilities align with realized outcomes across different low-probability bins.

The evaluation framework can also be extended with robustness checks such as a 10-cent threshold, a high-volume-only subset, a sport-specific split, and a time-to-cross analysis. These are not the core results of the paper, but they are natural follow-up checks that help verify whether the observed pattern is stable.

6 Discussion

These findings suggest that Kalshi sports markets exhibit strong longshot behavior and that low-price crossings are meaningful but not absolute indicators of final outcome. The fact that markets resolving NO crossed the 5-cent threshold more frequently than those resolving YES suggests that while crossing 5 cents is an indicator of extreme negative sentiment, it is not a definitive predictor of failure. This is consistent with the idea that prediction markets should be evaluated dynamically rather than only by their terminal prices.

This offers empirical evidence for the "Favorite-Longshot Bias" within regulated US prediction markets. In traditional sports betting, bettors often overvalue longshots, leading to poor calibration at the edges of the distribution. The "recovery" of markets that cross 5 cents and still resolve YES suggests that people who trade on Kalshi may overreact to transient, mid-game information. This drives prices into the longshot range even when the objective win probability remains higher than the implied price.

The results also highlight the difference between market pessimism and outcome certainty. A market that crosses below 5 cents is not necessarily doomed, but it has entered a region where beliefs are extremely negative. That makes first-crossing behavior a useful signal for studying calibration and market sentiment over time. However, the distribution in the min bid price in Table A2 suggests that liquidity acts as a significant stabilizing force. Higher-volume markets show smoother price paths, implying that some threshold crossings in thinner markets may be "false" signals triggered by temporary liquidity gaps rather than shifts in probability.

One important limitation is that the sample may include markets with very different liquidity levels. Thin markets may behave differently from high-volume markets, so the results should be interpreted with that in mind. Another limitation is that the exact threshold choice matters, which is why robustness checks are important. Despite these caveats, the results provide a strong empirical starting point for analyzing longshot behavior in prediction markets.

7 Conclusion

This project shows that first-crossing behavior is a useful way to study calibration and longshot dynamics in prediction markets. The analysis finds that markets resolving NO crossed below 5 cents more often than markets resolving YES, and that the distribution of minimum bid prices is heavily concentrated near zero. These results suggest that path-based market behavior contains important information that is missed by simple final-price evaluation.

More broadly, the project contributes a dynamic evaluation strategy for prediction markets by focusing on how prices evolve over time. Rather than treating market accuracy as a single final number, this approach examines the path that prices take before resolution. Future work could extend the analysis to other thresholds, other asset types, or additional market categories. Ultimately, this study shows that evaluating a market solely by its final price misses the full story. By tracking the 5-cent threshold as a "tail risk" zone, we can see how biases like the Favorite-Longshot Bias actually play out during active trading. This approach provides a blueprint for future research to separate real market shifts from temporary noise. As prediction markets become more popular and regulated, using these dynamic metrics will be key to understanding how the market will hold up under pressure.

8 Reference

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9 Appendix

9.1 Additional Files

The following files were generated during the project:

- `example_sports_series.csv`
- `market_outcomes_raw.csv`
- `market_outcomes.csv`
- `market_outcomes_highvol.csv`
- `highvol_tickers.txt`
- `first_crossing_results.csv`
- `bin_check_table.csv`
- `bootstrap_cis.csv`

9.2 Figure A5: Calibration Plot

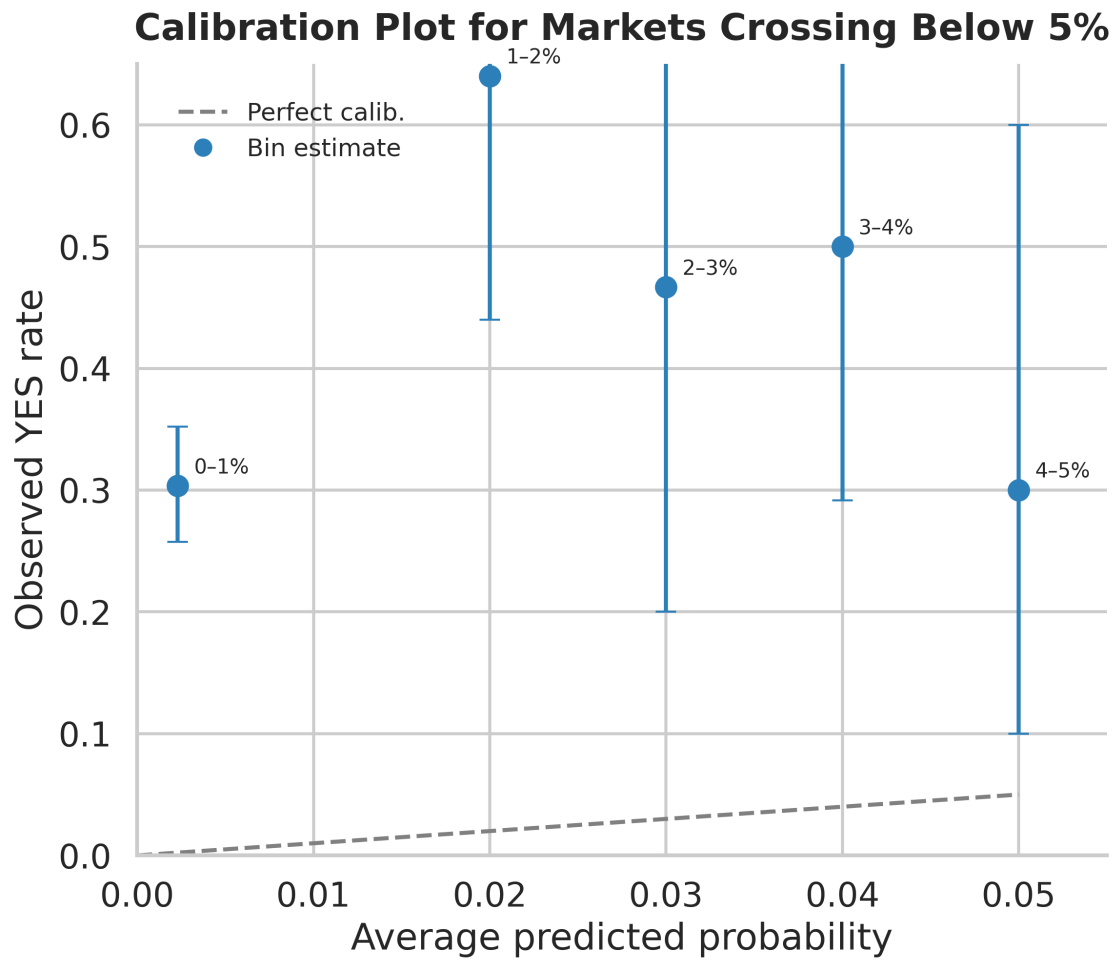


Figure 5: Calibration plot comparing average predicted probability to observed YES rate across low-probability bins.

9.3 Figure A6: Observed vs Predicted Probability by Bin

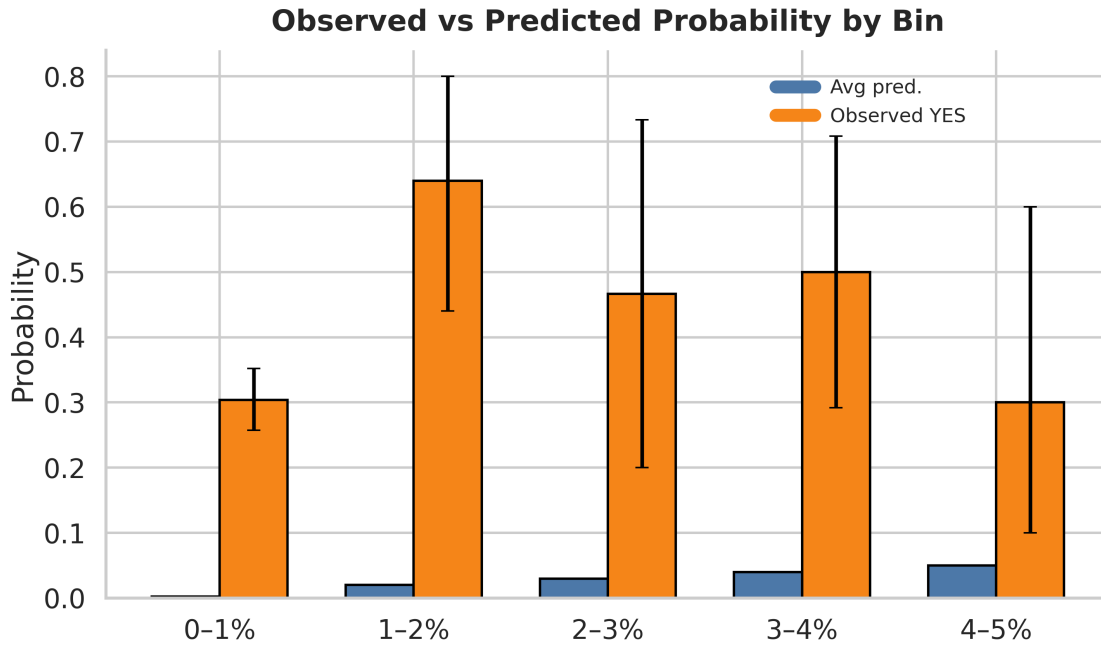


Figure 6: Observed and predicted probabilities by bin, with bootstrap confidence intervals on observed YES rates.

9.4 Table A1: Calibration Bin Results

Table 2: Bin-level calibration results and confidence intervals.

Bin	N	Avg. Predicted	Observed YES Rate	95% CI
0-1%	369	0.0023	0.3035	[0.2575, 0.3523]
1-2%	25	0.0200	0.6400	[0.4400, 0.8000]
2-3%	15	0.0300	0.4667	[0.2000, 0.7333]
3-4%	24	0.0400	0.5000	[0.2917, 0.7083]
4-5%	10	0.0500	0.3000	[0.1000, 0.6000]

9.5 Scraper Notes

The scraper queried the Kalshi API for sports series, market metadata, and candlestick data. Historical market candlesticks were downloaded at a 60-minute interval and stored separately by market ticker. Markets were filtered by resolution status and, in some versions of the analysis, by volume to reduce noise from thin trading.